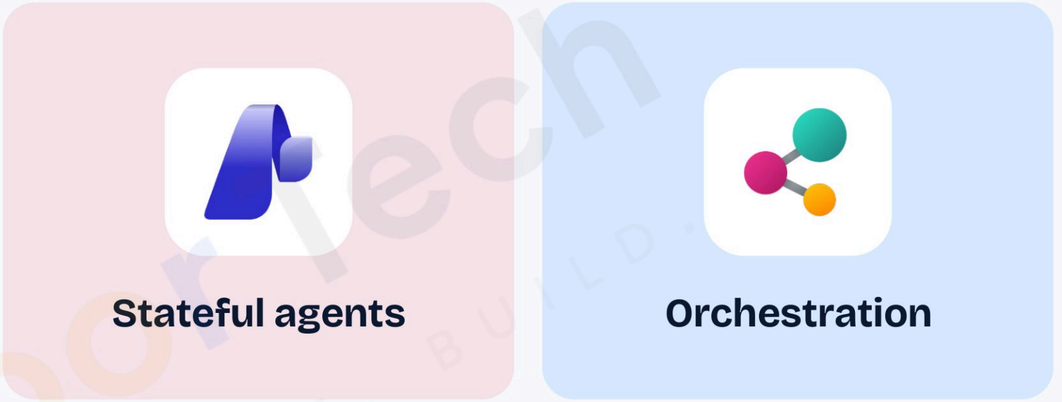
AI-500 COURSE INTRODUCTION

Design and implement multi-agent AI solutions on Azure

One insurer, four learning paths: from a single claims assistant to an enterprise of agents that plan, collaborate, ship and stay accountable.



Seventeen modules · Python · Microsoft Foundry and Agent Framework

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HELLO · THANK YOU FOR JOINING ME

Jalal Khan

Microsoft Certified Trainer

Training Manager, Project Manager & Solution Architect specializing in Power Platform, Conversational AI, Microsoft Azure, and Copilot Studio; delivering enterprise chatbot and automation solutions, leading teams, and enabling scalable AI adoption across industries.

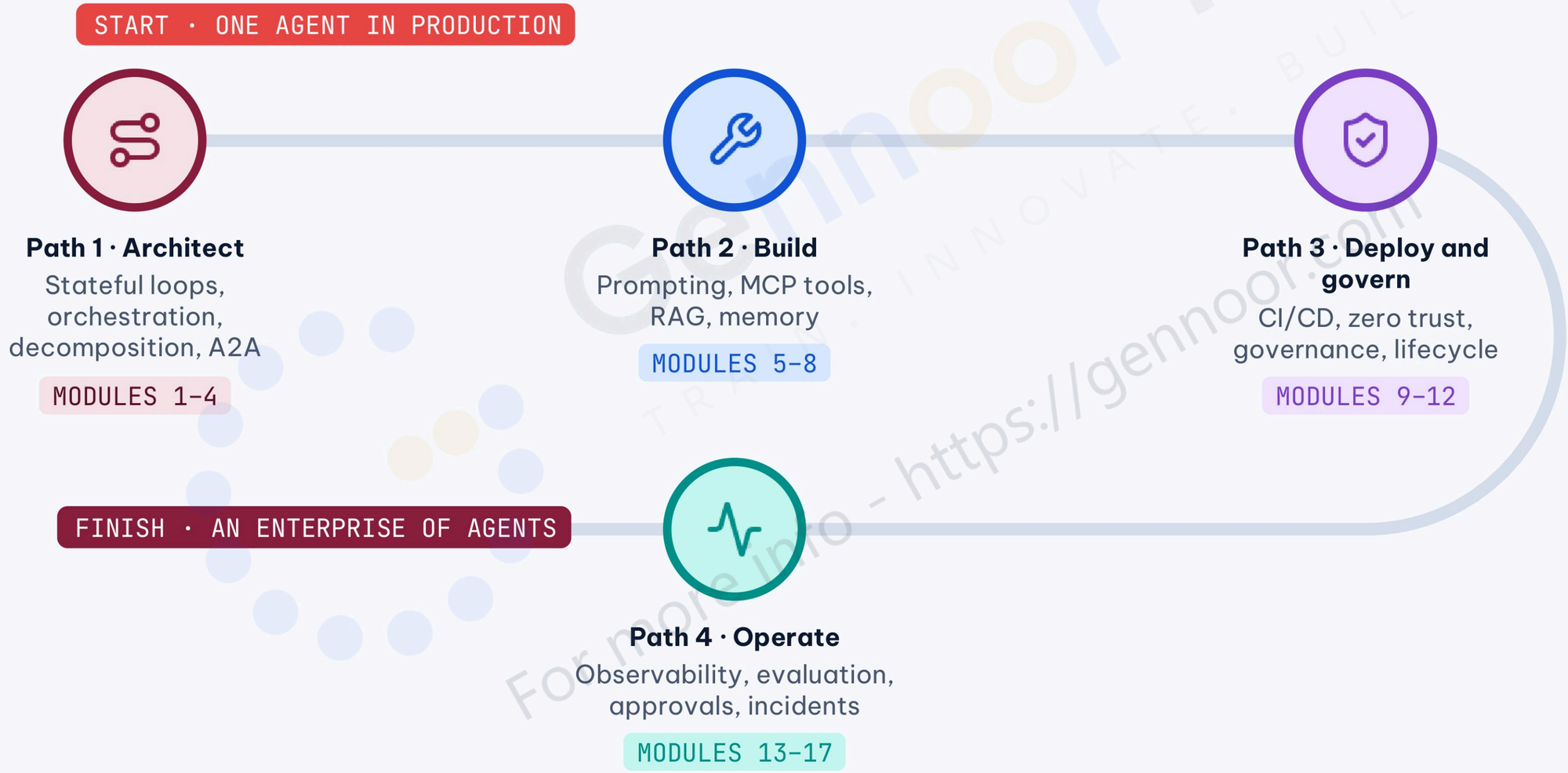
Delivered umpteen batches on Azure DevOps, Azure Development, Azure AI Engineer and customized batches on AI/ML.

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ONE STORY, FOUR LEARNING PATHS

Larkspur Mutual's claims assistant is live, and the business now asks for more than one agent can carry. Four learning paths take the team from one agent to a platform of agents.



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THE STORY

Meet Larkspur Mutual

An insurer with one agent in production and four problems you will recognize. Every section opens with a Larkspur chapter and closes with the Azure design.

LARKSPUR TODAY

One agent in production

Claims intake: answers policyholders, records a loss

Complex claims desk

A factory fire takes days of assessment, in several sessions

Partners and brokers

Partner agents to connect; broker data to keep apart

Money and regulators

Large payouts, exceptions, and "why was this decided?"

Current state, pressure, failure, decision, outcome: one chapter per section.

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0.1

Who this course is for

AI engineers and architects

You have built one agent and now design a system of agents: state, orchestration, tools, memory and operations.

Exam candidates

You are preparing for AI-500, the expert exam, and want the concepts, the code and the trade-offs it tests.

Teams running agents in production

You own agents in production and need release control, zero trust, governance and incident response.

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WHAT YOU BRING

Python with async · basic Azure: portal, CLI, resource groups · you have called a model API before. Foundry, agents, tools and the Agent Framework are introduced in this module.

0.2 Course map

Path 1 MODULES 1–4	**Architect production-grade multi-agent AI solutions in Azure**	M1 Stateful agentic loops · M2 Orchestration patterns · M3 Task decomposition and collaboration · M4 Agent communication with A2A
Path 2 MODULES 5–8	**Build production-grade multi-agent capabilities with Microsoft Foundry**	M5 Advanced prompting · M6 Tool ecosystems with MCP · M7 Advanced RAG with Azure AI Search · M8 Memory architectures with Cosmos DB
Path 3 MODULES 9–12	**Deploy and govern enterprise agentic AI solutions on Azure**	M9 CI/CD with GitHub Actions · M10 Zero-trust security · M11 Responsible AI governance with Content Safety · M12 Agent lifecycle governance
Path 4 MODULES 13–17	**Monitor, evaluate and operate multi-agent AI solutions in Azure**	M13 Observability with OpenTelemetry · M14 Evaluation frameworks · M15 Performance and cost · M16 Human-in-the-loop with Power Automate and Teams · M17 Production incidents

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0.3

How the exam is weighted

Skills measured, checked October 2026



EXAM Expert level · 700 of 1000 to pass · Python assumed · check the current skills outline on Microsoft Learn before you book

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0.4 Platform changes this course reflects

RETIRING	WHAT WE TEACH INSTEAD
Agents v1 (classic): threads, messages and runs RETIRES 31 MAR 2027	→ **Agents v2: conversations, responses and items with azure-ai-projects 2.x (Module 1)**
Foundry visual workflows in the portal RETIRE 1 DEC 2026	→ **Agent Framework workflows in code, deployed as hosted agents**
Prompt flow RETIRES 20 APR 2027	→ **Foundry evaluations and Agent Framework workflows (Module 14)**
Semantic Kernel and AutoGen agent packages SUCCESSOR ANNOUNCED OCT 2025	→ **Microsoft Agent Framework, the successor to both, used throughout**
Azure AI Foundry name and classic portal RENAMED NOV 2025	→ **Microsoft Foundry and the new portal experience, in every lab**

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0.5

How each section is taught

01 · STORY

Larkspur's chapter

Current state, pressure, failure, decision, outcome. Why the next topic matters.

02 · CONCEPT

Diagrams and trade-offs

A diagram that builds up, a For example flow, and a design decision with its costs.

03 · CODE

Current Python SDKs

azure-ai-projects 2.x, Agent Framework, Cosmos DB and OpenTelemetry, with today's names.

04 · LAB

One lab per module

Built in parts after each section. Printable guide, data and complete code.

05 · KEEP CURRENT

Retirements and successors

What is going away, when, and what to build on instead.

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0.6

Building on AI-103

AI-103 · ONE AGENT

Foundry
Resources, projects, model deployments

Agents
One agent, its instructions and its loop

Tools
Function tools, MCP, knowledge sources

RAG
Grounding answers in AI Search

Agent Framework basics
Agents and simple workflows in Python

AI-500 · A TEAM OF AGENTS IN PRODUCTION

Many agents, one outcome
Orchestration, planning, A2A · Path 1

Production-grade capabilities
Prompts, MCP tools, RAG, memory · Path 2

Deploy and govern
CI/CD, zero trust, responsible AI, lifecycle · Path 3

Operate
Tracing, evaluation, cost, people, incidents · Path 4

NEW TO AI-103? The next three primers cover what you need from it.

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0.7

Primer: Microsoft Foundry in one picture

PORTAL · ai.azure.com · NEW EXPERIENCE

Foundry resource

Billing and model deployments. One per team.

`larkspur-foundry`

Project

Your workspace: agents, connections, roles. One endpoint for all the code.

`claims`

Models

Deployments: gpt-5.4-mini by default, gpt-5.4 for reasoning, an embedding model

Agents

Named and versioned; Foundry Agent Service runs the loop for you

Tools

Function tools that run your Python, MCP servers, knowledge sources

Connections

Azure services the project may use: Cosmos DB, AI Search, Content Safety

FOR EXAMPLE · LARKSPUR'S SETUP

One resource, **larkspur-foundry** , in Sweden Central → One project, **claims** , with its endpoint in .env → **gpt-5.4-mini** and **gpt-5.4** deployed → The **claims-intake** agent, version 3, live

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0.8

Primer: what an agent is, and how it runs

Model
gpt-5.4-mini

+

Instructions
who it is, what it must do

+

Tools
functions, MCP, knowledge

=

Agent
A name and a version, stored in the project: `claims-intake v3`

THE FOUR-STEP LOOP · EVERY MODULE BUILDS ON IT

1 · YOU

Create a response on the conversation

"Fire at Vantor's Hamburg plant"

→

2 · AGENT

Answers with a function_call item

`get_policy("VM-2291")`

→

3 · YOUR CODE

Runs it, returns a function_call_output

the policy document, as text

→

4 · AGENT

More calls, or a message item

"Covered, section 4.2; reserve €180K"

↻

REPEAT

Until a message, or a budget says stop

Module 1, section 1

FOUR WORDS TO KNOW

`agent` A name and a version; created with the project client

`conversation` The history, kept by the service as items

`response` One step of work; sync, streaming or background

`item` message · function_call · function_call_output · reasoning

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0.9

Primer: the toolkit, and where each piece first appears



Microsoft Agent Framework

Agents and workflows in code; the successor to Semantic Kernel and AutoGen

MODULE 2



MCP

Model Context Protocol: the open standard linking agents to tools, local or remote

MODULE 6



A2A

Agent-to-agent protocol: agents publish a card, discover and call each other

MODULE 4



Azure Cosmos DB

State that survives: checkpoints, sessions, long-term memory

MODULES 1 · 8



Azure AI Search

Retrieval over policy wordings and documents: hybrid search and reranking

MODULE 7



Azure AI Content Safety

Screens inputs and outputs; the base of responsible AI governance

MODULE 11



Application Insights

OpenTelemetry traces: one trace per claim across the agent team

MODULE 13



GitHub Actions

Pipelines that test, evaluate and release agents, with rollback

MODULE 9

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UP NEXT · MODULE 1

Design stateful agentic loops

Larkspur's complex claims desk makes its claims agent durable: plan, critique, fork and resume, with checkpoints in Azure Cosmos DB.



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